



Fig. 5: Data acquisition of sample 1 through sample 3

load the AI model of your device and then start running on your Raspberry Pi computer. (See command.) Show the buttons to the camera connected to your Raspberry Pi, and it would count them.

Deploy model in Python

In the above deployment, it would work as intended in the Edge Impulse model. To make it work for your specific purpose, such as raising an audio alarm or lighting an LED when the count reaches + 2, you have to find some other means. Python3 comes to your aid. Now, Linux-sdk-python needs to be installed on your Raspberry Pi computer. Edge Impulse SDK (software development kit) is available for various models, such as Python, Node.js, C++, etc. The link below guides you to the page for SDK Python. (Link: <https://docs.edgeimpulse.com/docs/tools/edge-impulse-for-linux/linux-python-sdk>)

Once Linux-SDK-Python is installed, go to the linux-sdk-python/examples/image directory and run the Python file for image identification. In the example directory, there are three subdirectories—one each for audio data, image data, and custom data. In the image directory, the video classification file is also available for video input data. The custom directory is for customising other kinds of data (for experts' area only!).

```
$> python3 classify-image.py /home/bera/downloads/model.eim
```

That's how you load the Python file with the downloaded model.eim file. The program will automatically

find the camera module (USB connected, or Cam-Port connected) and will start running! On the top left corner, a small 120x120 camera window will open,

and the identified buttons will be marked with a small spot! The numbers identified will be shown on the terminal. Please ensure sufficient light is available, and the camera is properly focused on the buttons. This is particularly important for cheap cameras. That is why if you run the model on your smartphone, it produces far superior images and counts more quickly. Nonetheless, ensure proper light and focus; it will also produce satisfactory results.

Customise your model

Examine the classify-image.py file; it is a simple Python file that can be tailor-made with little understanding. In this Python file, we have added the espeak module so that the moment it finds the button(s), it speaks out the number of buttons it finds. Let's see part of the Python file to work around:

```
#!/usr/bin/env python
import device_patches # Device specific
patches - taken care of by the software
import cv2 #import Computer Vision
import os
import sys, getopt
import signal
import time
from edge_impulse_linux.image import
ImageImpulseRunner
import subprocess #this one has been
added by author.
elif "bounding_boxes" in res["result"].
keys():
```

```
print('Found %d bounding boxes (%d ms.)'
% (len(res["result"]["bounding_boxes"]),
res['timing']['dsp'] + res['timing']
['classification']))
if (len(res["result"]["bounding_
boxes"])>0):
    exitCode = subprocess.call(["espeak",
"-ven+f3", "-a200", " Found %d Buttons" %
len(res["result"]["bounding_boxes"]) ])
#This one has been added by author.
```

Espeak is a standalone text-to-speech module for Python. It does not require internet for working.

Modified run

Now you have modified the Python program. If you run the Python file now, it will locate the button [on the top left, a small 120x120 camera port will open], and the numbers will be shown on the terminal window, and the associated speaker will speak out the number—Found five buttons/ Found 2 buttons, etc. If you want to run some relay or light an LED, import the GPIO library of Python, and then fire the associated GPIO to run the relay. However, for running a relay, you have to use a switching transistor to increase the amount of current required for running the relay!

Aftermath

Edge Impulse computing started in 2019 with an objective to enable developers to create the next generation of intelligent devices. The AI-powered programs/devices started appearing on ESP32, Jetson Nano, Raspberry Pi, Orange Pi, Maixduino, OpenMV, Nicla Vision, and many more. This trend will further improve over the coming days! Gone are the days of supercomputers or big-brand, big-sized computers. Small, low-powered modular devices will cover that space in the future. **EFY**



Somnath Bera is an electronics enthusiast. He is a freelancer and has published several articles across the globe

EFY Note
The source code
of this project
is available

for free download at
www.electronicsforu.com